

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO3: Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)

Dave's Taxonomy: Precision (Level 3)

Question Count: 3

Total Marks: 30

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning	Total Marks
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case	
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application	
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis	
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution	
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized	

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytic sand demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

Introduction

A **Smart Retail Inventory Management System** is a cloud-based application used by retail stores to monitor products, stock levels, sales transactions, and customer demand. Store

devices such as **Point of Sale (POS) systems, barcode scanners, and computers** communicate with centralized cloud services through APIs.

Main Components

a. Clients

The clients are the devices used by employees and customers.

Examples:

- POS terminals
- Store computers
- Barcode scanners
- Manager dashboards
- Mobile devices

They send requests to cloud services through APIs.

API Gateway

The **API Gateway** acts as an entry point between clients and cloud services.

It provides:

- Request routing
- Authentication
- Access control
- API monitoring

Cloud Application Services

The application services process inventory-related operations.

Examples:

- Product management
- Stock management
- Sales processing
- Order management

Cloud Storage

Cloud storage stores:

- Product information

- Transaction records
- Product images

Security

Security is important because the system contains business and transaction information.

Security mechanisms include:

- **VPN:** Provides secure communication between stores and cloud services.
- **Firewall:** Blocks unauthorized network traffic.
- **API Gateway:** Controls access to APIs.
- **Authentication:** Verifies users.
- **Authorization:** Determines what each user can access.
- **Encryption:** Protects data during transmission and storage.

Smart Retail Inventory Management System

Cloud-Based Inventory Monitoring Dashboard

Total Products

5

Products

Total Stock

245

Units

Today's Sales

₹45,500

Revenue

Low Stock

2

Products

Add New Product

Product Name

Select Category

Price

Stock Quantity

Add Product

Inventory Details

ID	Product Name	Category	Price	Stock	Status
1	Laptop	Electronics	₹55,000	20	Available
2	Wireless Mouse	Accessories	₹800	50	Available
3	Keyboard	Accessories	₹1,200	7	Low Stock

Price

Stock Quantity

Add Product

Inventory Details

ID	Product Name	Category	Price	Stock	Status
1	Laptop	Electronics	₹55,000	20	Available
2	Wireless Mouse	Accessories	₹800	50	Available
3	Keyboard	Accessories	₹1,200	7	Low Stock
4	Smartphone	Electronics	₹25,000	35	Available
5	Headphones	Electronics	₹2,500	5	Low Stock
6	mobile	Electronics	₹25000	15	Available
7	smart mobile	Electronics	₹234	15	Available
8	USB Cable	Accessories	₹300	5	Low Stock

POS Transaction

Laptop

Quantity

8

USB Cable

This page says
Sale completed successfully!
Product: Laptop
Quantity: 2

5

Low Stock

POS Transaction

Laptop

Process Sale

Demand Forecasting

Product: Laptop

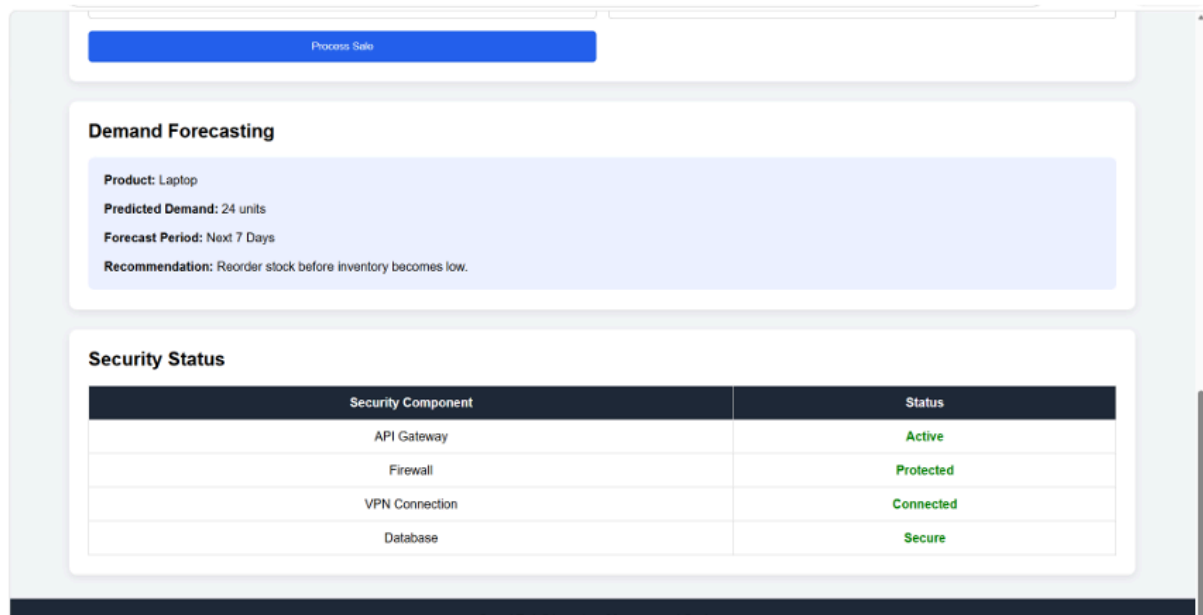
Predicted Demand: 24 units

Forecast Period: Next 7 Days

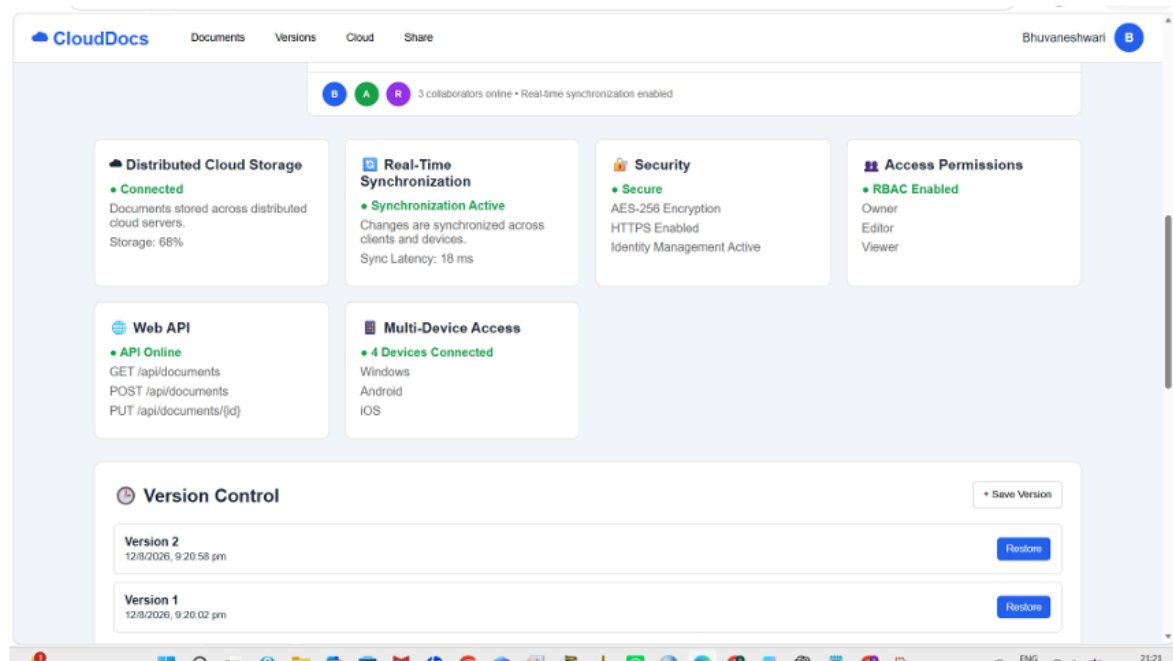
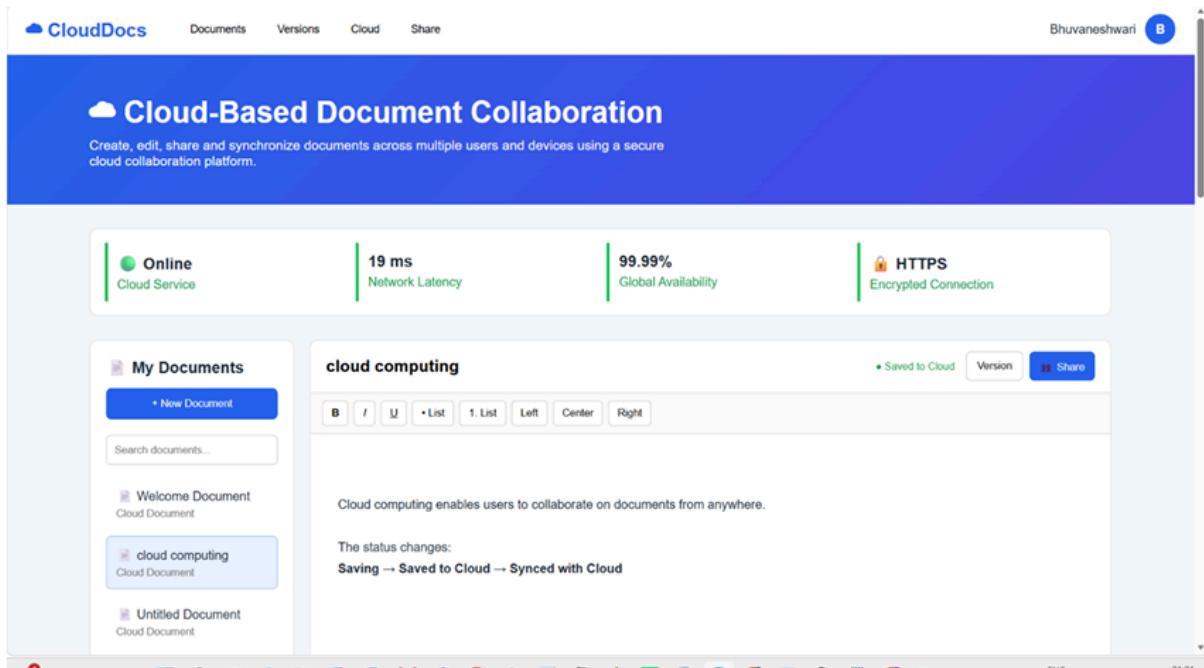
Recommendation: Reorder stock before inventory becomes low.

Security Status

Security Component	Status
API Gateway	Active
Firewall	Protected
VPN Connection	Connected



Case Study 2: Cloud-Based Document Collaboration System An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.



Cloud Platform Dashboard

Real-time cloud service monitoring

Total Documents
3Active Users
3API Requests
6129/minCloud Storage
68%Network Latency
24 msAvailability
99.99%

Recent Activity

- 9:21:39 pm - Opened document: cloud computing
- 9:21:37 pm - Opened document: Welcome Document
- 9:20:58 pm - New document version created.
- 9:20:43 pm - Document shared with bhuvana@gmail.com as Editor
- 9:20:02 pm - New document version created.
- 9:20:01 pm - Document saved to distributed cloud storage.

Recent Activity

- 9:21:39 pm - Opened document: cloud computing
- 9:21:37 pm - Opened document: Welcome Document
- 9:20:58 pm - New document version created.
- 9:20:43 pm - Document shared with bhuvana@gmail.com as Editor
- 9:20:02 pm - New document version created.
- 9:20:01 pm - Document saved to distributed cloud storage.
- 9:19:35 pm - Document synchronized across connected clients.
- 9:19:35 pm - Document saved to distributed cloud storage.

CloudDocs

Cloud-Based Document Collaboration
System

Collaboration

Real-Time Editing
Document Sharing
Multi-Device Synchronization

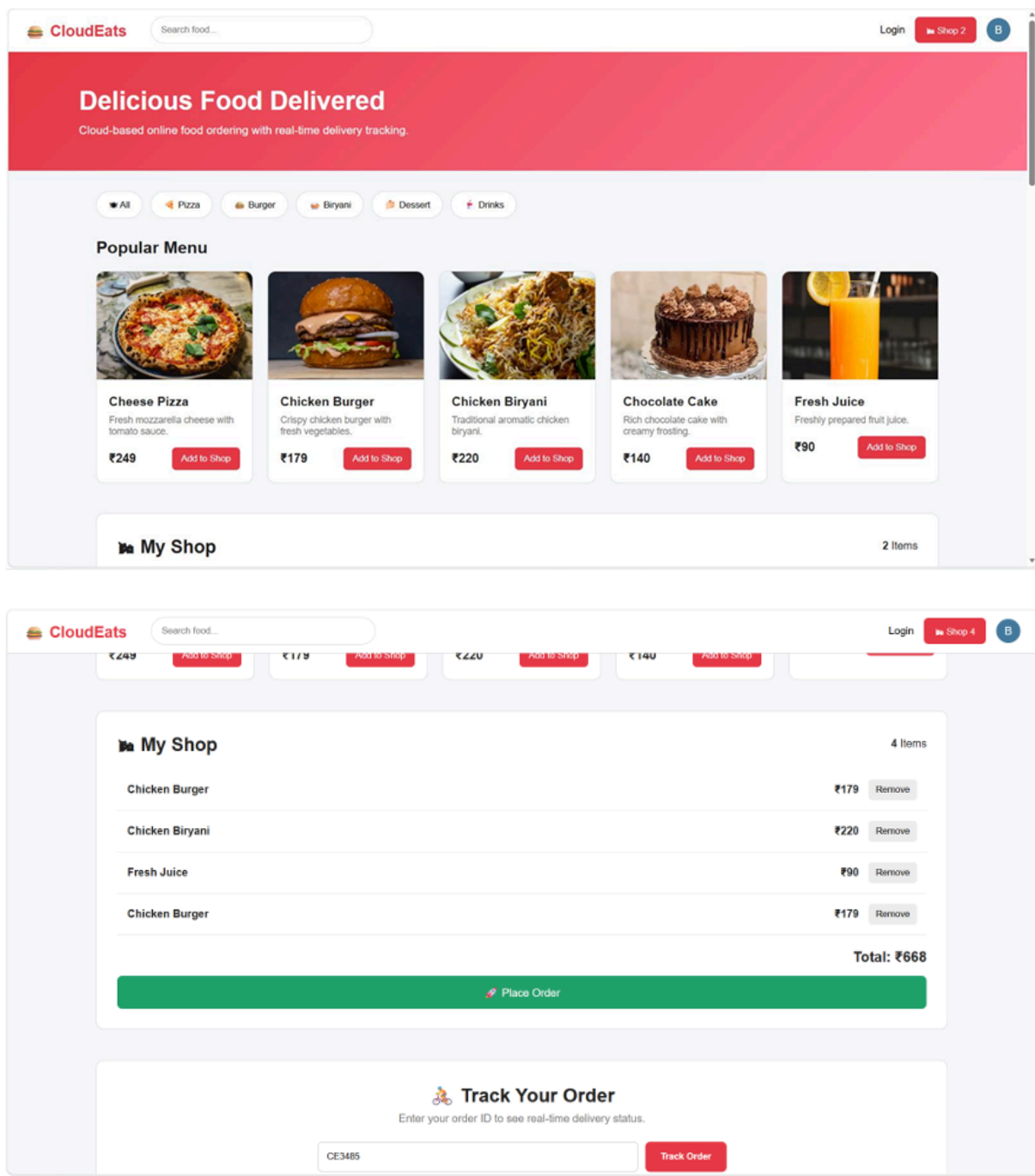
Cloud Services

Distributed Storage
Web APIs
Version Control

Security

HTTPS Encryption
Identity Management
Access Permissions

Case Study 3: Online Food Ordering Platform A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.



Cloud Infrastructure

REST API

• API Online

GET /api/menu
POST /api/orders
GET /api/orders/{id}

Virtual Machines

• 4 VMs Running

CPU: 42%
Memory: 58%

Kubernetes

• Cluster Healthy

Pods: 12 Running
Auto Scaling Enabled

Cloud Storage

• Connected

Menu Images: 245
Invoices: 1,248

Security

• Secure

HTTPS Enabled
Token Authentication

RBAC

• Enabled

Customer
Restaurant Admin
Delivery Agent

Cloud Platform Dashboard

Real-time cloud service monitoring

Total Orders

2

Active Users

1

API Requests

8342/min

Availability

99.99%